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Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 22	
TITLE : JDBC PreparedStatement, and ResultSet	

Problem Statement

Demonstration and exploration of JDBC classes and interfaces (PreparedStatement, and ResultSet).

Learning Objectives

To implement PreparedStatement and ResultSet for secure query execution and data retrieval.

Programme

1: Create a login application using PreparedStatement to check username and password.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

public class LoginApplication {

    public static void main(String[] args) throws Exception {

        Class.forName("com.mysql.cj.jdbc.Driver");

        String db = "jdbc:mysql://localhost:3306/employee";
        String user = "root";
        String password = "admin@123";

        try {
            Connection con = DriverManager.getConnection(db,
user, password);

            Scanner sc = new Scanner(System.in);
```

```

        System.out.print("Enter username: ");
        String username = sc.nextLine();

        System.out.print("Enter password: ");
        String pass = sc.nextLine();

        String query = "SELECT * FROM login WHERE username
= ? AND password = ?";

        PreparedStatement ps =
con.prepareStatement(query);

        ps.setString(1, username);
        ps.setString(2, pass);

        ResultSet rs = ps.executeQuery();

        if (rs.next()) {
            System.out.println("Login successful");
        } else {
            System.out.println("Invalid username or
password");
        }

        rs.close();
        ps.close();
        con.close();
        sc.close();

    } catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
    }
}
}

```

```

piyushrohokale@Piyushs-MacBook-Air JDBC % java -cp "src:lib/mysql-connector-j-26.7.0.jar" LoginApplication
Enter username: admin
Enter password: 123
Invalid username or password
piyushrohokale@Piyushs-MacBook-Air JDBC % java -cp "src:lib/mysql-connector-j-26.7.0.jar" LoginApplication
Enter username: admin
Enter password: 1234
Login successful
piyushrohokale@Piyushs-MacBook-Air JDBC %

```

2: Create a Java application using PreparedStatement to authenticate hospital staff login (doctor/nurse/login ID and password). Display appropriate access messages based on authentication result.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

public class HospitalStaffLogin {

    public static void main(String[] args) throws Exception {

        Class.forName("com.mysql.cj.jdbc.Driver");

        String db = "jdbc:mysql://localhost:3306/employee";
        String user = "root";
        String password = "admin@123";

        try {
            Connection con = DriverManager.getConnection(db,
user, password);

            Scanner sc = new Scanner(System.in);

            System.out.print("Enter staff type (Doctor/Nurse):
");
            String staffType = sc.nextLine();

            System.out.print("Enter login ID: ");
            String loginId = sc.nextLine();

            System.out.print("Enter password: ");
            String pass = sc.nextLine();

            String query = "SELECT * FROM hospital_staff WHERE
staff_type = ? AND login_id = ? AND password = ?";

            PreparedStatement ps =
con.prepareStatement(query);

            ps.setString(1, staffType);
            ps.setString(2, loginId);
            ps.setString(3, pass);

            ResultSet rs = ps.executeQuery();

            if (rs.next()) {
                if (staffType.equalsIgnoreCase("Doctor")) {
```

```

        System.out.println("Doctor login
successful");
        System.out.println("Access granted to
doctor portal");
    } else if
(staffType.equalsIgnoreCase("Nurse")) {
        System.out.println("Nurse login
successful");
        System.out.println("Access granted to
nurse portal");
    }
    } else {
        System.out.println("Authentication failed");
        System.out.println("Invalid staff type, login
ID or password");
    }
}

rs.close();
ps.close();
con.close();
sc.close();

} catch (Exception e) {
    System.out.println("Error: " + e.getMessage());
}
}
}
}

```

```

piyushrohokale@Piyushs-MacBook-Air JDBC % java -cp "src:lib/mysql-connector-j-26.7.0.jar" HospitalStaffLogin
Enter staff type (Doctor/Nurse): Nurse
Enter login ID: Nurse01
Enter password: Nurse123
Nurse login successful
Access granted to nurse portal
piyushrohokale@Piyushs-MacBook-Air JDBC % 

```

Conclusion

The program successfully demonstrates hospital staff authentication using Java JDBC and PreparedStatement. It verifies the staff type, login ID, and password from the database and displays appropriate access messages for doctors and nurses based on the authentication result.

Github:

Java_assignments / assignment22 /

Add file ...

 VyankateshRohokale

assignment 22

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 HospitalStaffLogin.java	assignment 22	now
 LoginApplication.java	assignment 22	2 minutes ago